

Pattern-Calibrated Multimodal Prediction under Blockwise Missingness

Junhan Yu^{1,*}, Kejian Zhang^{1,*}, Doudou Zhou^{1,*,†}, Guojun Zhu^{2,*}

¹ Department of Statistics & Data Science, National University of Singapore

² School of Mathematical Sciences, University of Chinese Academy of Sciences

* Alphabetical order

† Corresponding author: ddzhou@nus.edu.sg

Abstract

Blockwise missingness in multimodal data creates observed-modality patterns with different prediction targets. A pooled rule need not be optimal for any fixed pattern, whereas pattern-specific fitting can be inefficient when some patterns are rare. Borrowing is especially difficult when two patterns share no raw modality and hence no directly comparable covariate component. We propose Multimodal Overlap-aware Shared-specific Alignment and Inter-pattern Calibration, a framework that learns aligned shared and modality-specific representations from partially observed multimodal data, then estimates a separate predictor for each pattern. For a focal pattern, observations from all patterns contribute only through representation components available under the focal pattern; focal-pattern observations then estimate the remaining correction. Under a linear latent-representation model, non-asymptotic bounds show how representation coverage, correction size, and representation error determine whether borrowing improves on local fitting. They also show that, when co-observed data align the modalities upstream, observations from raw-disjoint patterns can increase the effective sample size for estimating the shared component. Simulations, a naturally incomplete intensive-care mortality application, and controlled-masking studies of emotion recognition and glaucoma classification show that the method is most useful when focal-pattern samples are limited and pattern-specific discrepancies are appreciable.

Keywords: Blockwise missingness; Multimodal prediction; Pattern-specific prediction; Cross-pattern borrowing; Shared-specific representation learning.

1 Introduction

Prediction studies increasingly combine heterogeneous modalities, including structured records, text, images, and molecular assays (Baltrušaitis et al. 2019, Kline et al. 2022, Argelaguet et al. 2018). Because these modalities arise from different acquisition systems and workflows, only a subset may be observed for each subject; in electronic health records (EHRs), for example, structured codes, clinical notes, and chest images may be available for different subsets of patients (Krones et al. 2025). This blockwise missingness induces observed-modality patterns, each defining the information available for prediction and therefore potentially inducing a different optimal prediction rule (Fletcher Mercaldo & Blume 2020, Le Morvan, Prost, Josse, Scornet & Varoquaux 2020). A common pooled rule across patterns may therefore fail to recover the optimum for a given pattern, whereas fitting each pattern separately can be inefficient when pattern-specific samples are small. Figure 1 illustrates these pattern-specific prediction tasks.

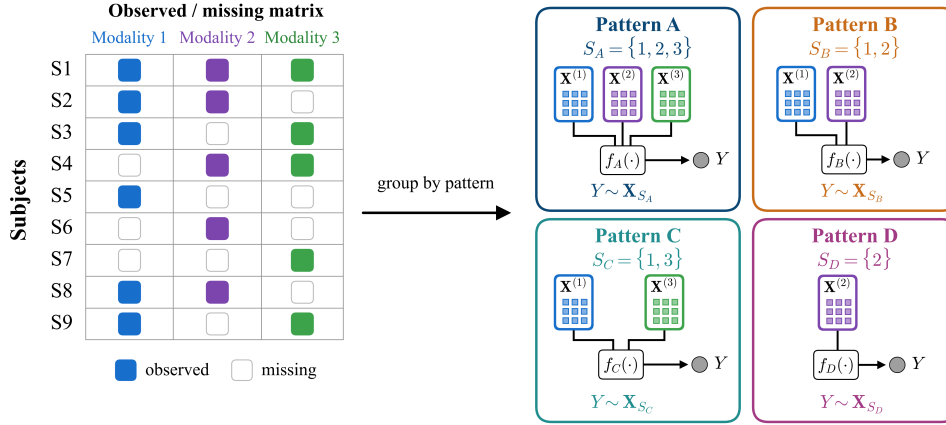


Figure 1: Blockwise missingness induces observed-modality patterns with different information sets. Each pattern defines the covariates available to its prediction rule.

Prior work has addressed pattern-dependent prediction through local fitting, mask-dependent modelling, and parameter sharing. Pattern submodels fit a separate prediction rule for each missingness pattern using only observations from that pattern (Fletcher Mercaldo & Blume

2020). Mask-dependent Bayes prediction has been represented through interactions between observed values and missingness indicators, mask-aware neural architectures, and learned representations of the missingness pattern (Le Morvan, Prost, Josse, Scornet & Varoquaux 2020, Le Morvan, Josse, Moreau, Scornet & Varoquaux 2020, Ma et al. 2025). Sharing pattern submodels (SPSM) couple a common coefficient vector with sparse pattern-specific deviations, allowing information sharing without forcing every pattern to use the same rule (Stempfle et al. 2023). However, all three strategies express the pattern-specific predictors in a common covariate system.

A complementary line of work extracts predictive information from incomplete observations. Multimodal methods handle missing modalities by reconstructing them or by using zero-filling, missingness masks, reconstruction losses, and specialized fusion architectures (Tran et al. 2017, Chartsias et al. 2018, Wu & Goodman 2018, Shi et al. 2019, Yuan et al. 2021, Zuo et al. 2023, Wu et al. 2026). Related shared-specific methods separate representations common across modalities from those specific to each modality (Hazarika et al. 2020, Lee & Pavlovic 2021, Wang et al. 2023). Statistical methods borrow through prespecified structures: HAM uses observations containing the covariate subsets required by particular ANOVA components (Sell et al. 2024), DISCOM pools moments across available modality pairs (Yu et al. 2020), and Modular Regression uses designated auxiliary covariates or tasks (Jin & Rothenhäusler 2023). Thus, these approaches provide either multimodal fusion and representation learning or borrowing through predefined covariate or task relationships. What remains unresolved is how to construct comparable representations for heterogeneous observed-modality patterns and borrow across them while retaining a distinct prediction rule for each pattern.

Extending pattern-aware borrowing to multimodal data therefore requires solving two linked problems. First, two patterns may share no raw covariates. In EHR data, for example, a

code-only pattern and a note-only pattern provide no common covariate through which information can be borrowed. Borrowing between them therefore requires representations that make information extracted from different modalities comparable. Such representations can be learned from observations in which the relevant modalities are jointly observed, even when the two patterns themselves have no raw modality in common. Second, borrowing must preserve the prediction rule of every pattern rather than produce one global multimodal predictor. For a focal pattern, observations from another pattern can contribute only through representation components available under both patterns. Observations from the focal pattern must then be used to estimate the gap between the resulting preliminary predictor and the focal pattern’s prediction rule.

To address these two problems, we propose **Multimodal Overlap-aware Shared-specific Alignment and Inter-pattern Calibration (MOSAIC)**. MOSAIC first learns aligned shared and modality-specific representations from partially observed multimodal observations without using response labels. For each pattern, it then forms a preliminary predictor using observations across all patterns, while retaining from each pattern only the representation components also available under the focal pattern. MOSAIC uses observations from the focal pattern to calibrate the preliminary predictor to that pattern’s prediction rule. The preliminary predictor can therefore be estimated using more observations than a local fit, while the final predictor remains pattern-specific.

Our contributions are three-fold. First, we formulate blockwise multimodal prediction as a collection of pattern-specific risk minimization problems over potentially disjoint raw covariate spaces, thereby distinguishing each pattern’s target from the average risk optimized by a pooled predictor. Second, we develop a unified estimation procedure that operates on modality-specific inputs and uses observations across patterns to estimate a distinct predictor for each pattern. Third, our non-asymptotic analysis characterizes how

representation coverage, the gap between the preliminary predictor and the focal-pattern rule, and representation estimation error determine the benefit of borrowing. It identifies regimes in which MOSAIC improves on local fitting and shows that observations from patterns with no raw modality in common can increase the effective sample size for estimating the shared component of a pattern-specific predictor. Simulations and multimodal applications examine the roles of representation learning, cross-pattern borrowing, and pattern-specific calibration.

The rest of the paper is organized as follows. Section 2 presents the proposed method. Section 3 establishes theoretical guarantees. Section 4 reports simulation results; Section 5 presents real-data analyses; and Section 6 concludes.

2 Methodology

2.1 Problem Setup and Method Overview

For subject i , let y_i be the response and let $\mathbf{x}_i^{(l)}$ denote the raw input from modality $l \in [L]$, where $[L] = \{1, \dots, L\}$. For any modality set $A \subseteq [L]$, write $\mathcal{X}_i(A) = \{\mathbf{x}_i^{(l)} : l \in A\}$, and let $\mathcal{X}(A)$ denote the corresponding generic multimodal observation. The complete multimodal observation $\mathcal{X}_i([L])$ may contain heterogeneous objects such as structured codes, text, or images. Let $m_i^{(l)} \in \{0, 1\}$ indicate whether modality l is observed, and define $\mathcal{A}_i = \{l \in [L] : m_i^{(l)} = 1\}$, with $\mathcal{A}$ denoting the corresponding generic random set. The realized set $\mathcal{A}_i$ is subject i 's observed-modality pattern; we denote K distinct nonempty values of $\mathcal{A}$ by $A_1, \dots, A_K$.

Because the available covariates differ by pattern, the prediction target must first be defined for each pattern. Let $\mathfrak{X}_{A_k}$ denote the raw covariate space for observations with modalities

A_k . For A_k , define the conditional risk and its raw-space minimizer as

$$R_k^{\text{raw}}(g) := \mathbb{E} \left[\{Y - g(\mathcal{X}(A_k))\}^2 \mid \mathcal{A} = A_k \right], \quad g_{(k)}^{\text{raw}} := \arg \min_{g \in \mathcal{G}_k^{\text{raw}}} R_k^{\text{raw}}(g), \quad (2.1)$$

where $\mathcal{G}_k^{\text{raw}}$ is a class of prediction rules using the raw modalities in pattern A_k . This is the population target of pattern submodels (Fletcher Mercaldo & Blume 2020); when $\mathcal{G}_k^{\text{raw}}$ is unrestricted, it is the Bayes predictor conditional on the observed values and missingness pattern (Le Morvan, Prost, Josse, Scornet & Varoquaux 2020).

To contrast this target with pooling, let $\tilde{\mathcal{G}}^{\text{raw}}$ be a class of predictors $\tilde{g}(x, A)$, defined for $A \in \{A_1, \dots, A_K\}$ and $x \in \mathfrak{X}_A$. For every $\tilde{g} \in \tilde{\mathcal{G}}^{\text{raw}}$ and $k \in [K]$, let $\tilde{g}_k(x) := \tilde{g}(x, A_k)$ and suppose $\tilde{g}_k \in \mathcal{G}_k^{\text{raw}}$. The pooled population rule is

$$\begin{aligned} \tilde{g}_{\text{pool}}^{\text{raw}} &:= \arg \min_{\tilde{g} \in \tilde{\mathcal{G}}^{\text{raw}}} \mathbb{E} \left[\{Y - \tilde{g}(\mathcal{X}(\mathcal{A}), \mathcal{A})\}^2 \right] \\ &= \arg \min_{\tilde{g} \in \tilde{\mathcal{G}}^{\text{raw}}} \sum_{k=1}^K \pi_k \mathbb{E} \left[\{Y - \tilde{g}(\mathcal{X}(A_k), A_k)\}^2 \mid \mathcal{A} = A_k \right], \end{aligned}$$

where $\pi_k = \Pr(\mathcal{A} = A_k)$.

Proposition 1 (Pattern-specific excess risk of pooled rules) *Suppose that, for each $k \in [K]$, $\mathbb{E}(Y^2 \mid \mathcal{A} = A_k) < \infty$ and $\mathcal{G}_k^{\text{raw}}$ is closed and convex in $L_2(P_k)$, where P_k is the conditional distribution of $\mathcal{X}(A_k)$ given $\mathcal{A} = A_k$. Then, for the pooled population rule $\tilde{g}_{\text{pool}}^{\text{raw}}$,*

$$R_k^{\text{raw}}(\tilde{g}_{\text{pool},k}^{\text{raw}}) - R_k^{\text{raw}}(g_{(k)}^{\text{raw}}) \geq \left\| \tilde{g}_{\text{pool},k}^{\text{raw}} - g_{(k)}^{\text{raw}} \right\|_{L_2(P_k)}^2, \quad \forall k \in [K].$$

Proposition 1 identifies the approximation cost of pooling: whenever the rule induced by the pooled model differs from $g_{(k)}^{\text{raw}}$, pattern A_k incurs positive excess risk. If the pooled class can realize all K pattern-specific optima simultaneously, this cost vanishes. Let n_k denote the number of labelled observations with pattern A_k in a sample of size $n = \sum_{k=1}^K n_k$. A pattern-saturated class can attain all K optima, but its empirical risk minimization separates into pattern-specific fits, each based only on n_k observations. Imposing cross-pattern structure can reduce estimation error when some n_k are small, but introduces approximation bias if

the structure excludes any pattern-specific optimum. Useful borrowing must therefore link the pattern-specific rules without forcing them to coincide.

In multimodal data, constructing such a link is difficult because comparable predictor components may not exist in the raw covariate space. If $A_j \cap A_k = \emptyset$, observations from the two patterns contain different raw variables, so their raw-space predictors have no covariate-indexed coefficient or function component that can be matched and shared directly. Shared latent representations provide a way around this limitation by mapping different modalities into comparable representations (Lee & Pavlovic 2021, Wang et al. 2023). These works, however, do not study how such representations can support statistical borrowing across pattern-specific prediction rules.

To enable such borrowing, MOSAIC constructs a common latent coordinate system to represent the complete multimodal information. Formally, the complete multimodal information is represented by a shared representation $\mathbf{z}_i^{(0)} \in \mathbb{R}^{d^{(0)}}$ and modality-specific representations $\mathbf{z}_i^{(l)} \in \mathbb{R}^{d^{(l)}}$ for $l \in [L]$. The raw modalities are generated from these representations as

$$\mathbf{x}_i^{(l)} = f_x^{(l)}(\mathbf{z}_i^{(0)}, \mathbf{z}_i^{(l)}, \boldsymbol{\varepsilon}_{x,i}^{(l)}), \quad l \in [L], \quad \mathbf{z}_i = (\mathbf{z}_i^{(0)\top}, \mathbf{z}_i^{(1)\top}, \dots, \mathbf{z}_i^{(L)\top})^\top \in \mathbb{R}^d, \quad (2.2)$$

where the $f_x^{(l)}$ are unknown functions, $\boldsymbol{\varepsilon}_{x,i}^{(l)}$ is the noise, and $d = d^{(0)} + \sum_{l=1}^L d^{(l)}$. Although $\mathbf{z}_i$ in (2.2) contains representation blocks associated with all modalities, an observation with pattern A_k provides only the shared block $\mathbf{z}_i^{(0)}$ and the modality-specific blocks $\{\mathbf{z}_i^{(l)} : l \in A_k\}$. We therefore index the representations available under A_k by $S_k = \{0\} \cup A_k$, where 0 denotes the shared representation, and write their total dimension as $d_k = \sum_{l \in S_k} d^{(l)}$.

This alignment changes what can be borrowed. For a focal pattern A_k , every observation supplies an estimate of the same aligned shared block $\mathbf{z}^{(0)}$, regardless of which raw modalities produced it. An observation with pattern A_j additionally supplies a modality-specific block l required under A_k only when $l \in A_j \cap A_k$. Consequently, even when $A_j \cap A_k = \emptyset$, observations with pattern A_j still enter the preliminary prediction step through $\mathbf{z}^{(0)}$, which

acts as a channel for information sharing despite the absence of common raw modalities. This distinction is formalized in Proposition 3. Unlike prior work that shares coefficients through common raw modalities, MOSAIC can borrow through the shared representation even when two patterns have disjoint raw modalities.

The aligned representations provide fixed-dimensional inputs for standard prediction algorithms as well as the coordinates needed for cross-pattern borrowing. For pattern A_k , let $\mathbf{z}_{(k)} \in \mathbb{R}^{d_k}$ denote the concatenation of the representations indexed by S_k , and let $\mathcal{G}_k$ be a class of prediction functions from $\mathbb{R}^{d_k}$ to $\mathbb{R}$. We define the representation-space target as

$$g_{(k)} := \arg \min_{g \in \mathcal{G}_k} \mathbb{E} \left[\{Y - g(\mathbf{z}_{(k)})\}^2 \mid \mathcal{A} = A_k \right]. \quad (2.3)$$

MOSAIC estimates a separate $g_{(k)}$ for each pattern. For a focal pattern A_k , it first uses observations from all patterns to estimate a preliminary predictor h_k from the representation blocks that each pattern can supply. It then uses observations with pattern A_k to estimate the remaining discrepancy. At the population level,

$$g_{(k)} = h_k + \Delta_k, \quad \Delta_k := g_{(k)} - h_k.$$

Borrowing is beneficial when observations across patterns provide sufficient coverage to estimate h_k accurately and Δ_k is easier to estimate from the n_k focal-pattern observations than the full rule $g_{(k)}$. Figure 2 summarizes these three stages of MOSAIC; the next three subsections define the representation learner, preliminary predictor, and focal-pattern calibration.

2.2 Learning Representations for Cross-Pattern Borrowing

We now describe how MOSAIC learns the aligned shared and modality-specific representations used for cross-pattern borrowing.

The representation-learning sample consists of N multimodal observations indexed by

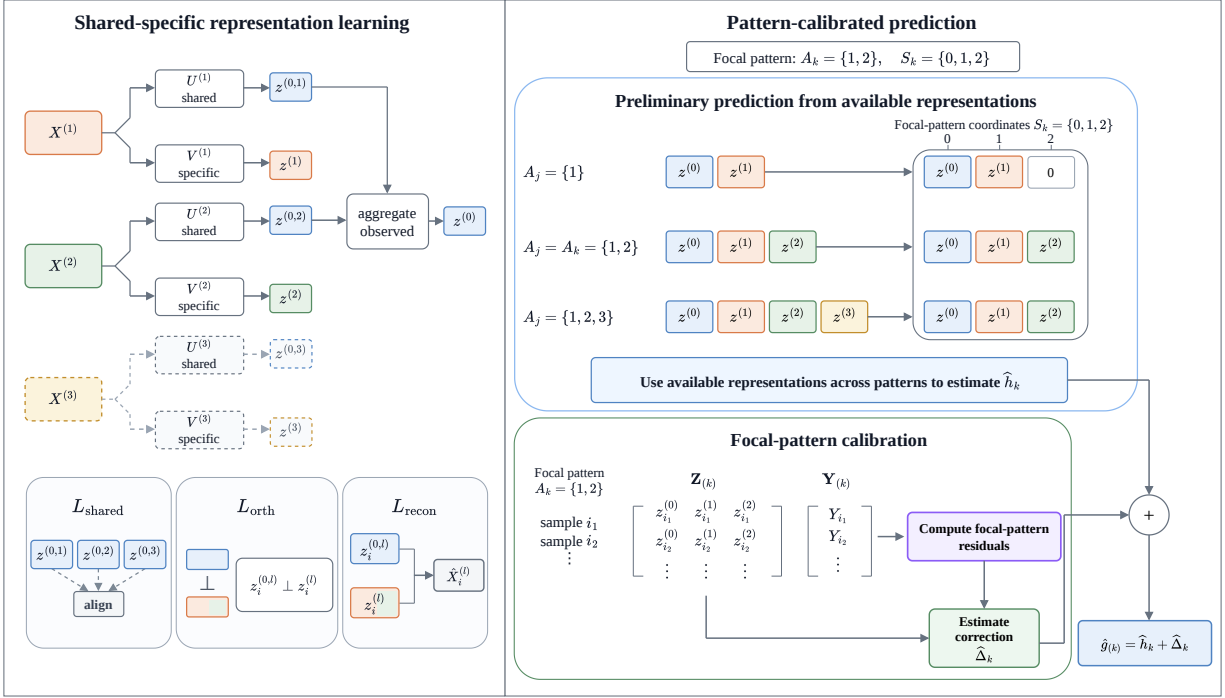


Figure 2: Overview of MOSAIC in a three-modality example with focal pattern $A_k = \{1, 2\}$ and representation index set $S_k = \{0, 1, 2\}$. MOSAIC learns aligned shared and modality-specific representations, constructs a preliminary predictor by retaining from each pattern only the representation components available under the focal pattern, and estimates the remaining correction from focal-pattern observations.

$i = n + 1, \dots, n + N$, whose response labels are not used and may be unavailable. This separates it from the labelled prediction sample indexed by $i = 1, \dots, n$. After modality-specific preprocessing, each observed modality is represented as a vector in $\mathbb{R}^{d_{\text{raw}}^{(l)}}$. For each modality $l \in [L]$, we learn a shared representation encoder $U^{(l)} : \mathbb{R}^{d_{\text{raw}}^{(l)}} \rightarrow \mathbb{R}^{d^{(0)}}$ and a modality-specific representation encoder $V^{(l)} : \mathbb{R}^{d_{\text{raw}}^{(l)}} \rightarrow \mathbb{R}^{d^{(l)}}$.

The shared encoders are aligned using observations in which two modalities are jointly observed. Since the shared representation participates in the generation of every modality, their shared encoder outputs for the same subject should be close. For pairs of distinct

modalities with positive co-observation counts, we use:

$$\mathcal{L}_{\text{shared}} = \frac{1}{|\mathcal{P}_N|} \sum_{(l,l') \in \mathcal{P}_N} \frac{1}{\sum_{i=n+1}^{n+N} m_i^{(l)} m_i^{(l')}} \sum_{i=n+1}^{n+N} m_i^{(l)} m_i^{(l')} \left[\ell_{\text{align}} \left(U^{(l)}(\mathbf{x}_i^{(l)}), U^{(l')}(\mathbf{x}_i^{(l')}) \right) \right], \quad (2.4)$$

where $\mathcal{P}_N = \{(l, l') : 1 \leq l < l' \leq L, \sum_{i=n+1}^{n+N} m_i^{(l)} m_i^{(l')} > 0\}$ and ℓ_{align} can be the squared Euclidean distance or a contrastive alignment loss. We next regularize the empirical second moment of the concatenated representation toward the identity, which fixes its scale and discourages linear dependence both within and between the shared and modality-specific blocks. Define $W^{(l)}(\mathbf{x}_i^{(l)}) = \left[U^{(l)}(\mathbf{x}_i^{(l)})^\top, V^{(l)}(\mathbf{x}_i^{(l)})^\top \right]^\top$ and let $\|\cdot\|_F$ denote the Frobenius norm. We impose:

$$\mathcal{L}_{\text{orth}} = \frac{1}{L} \sum_{l=1}^L \left\| \frac{1}{\sum_{i=n+1}^{n+N} m_i^{(l)}} \sum_{i=n+1}^{n+N} m_i^{(l)} W^{(l)}(\mathbf{x}_i^{(l)}) W^{(l)}(\mathbf{x}_i^{(l)})^\top - \mathbf{I}_{d^{(0)}+d^{(l)}} \right\|_F^2. \quad (2.5)$$

However, orthogonality alone does not ensure that the modality-specific representation captures meaningful modality-specific information; it may instead learn arbitrary orthogonal noise. To prevent this, the representations are also required to retain information needed to reconstruct the original modality. With a decoder $T^{(l)}$, we use:

$$\mathcal{L}_{\text{recon}} = \frac{1}{L} \sum_{l=1}^L \frac{1}{\sum_{i=n+1}^{n+N} m_i^{(l)}} \sum_{i=n+1}^{n+N} m_i^{(l)} \left\| \mathbf{x}_i^{(l)} - T^{(l)} \left(\left[U^{(l)}(\mathbf{x}_i^{(l)}); V^{(l)}(\mathbf{x}_i^{(l)}) \right] \right) \right\|_2^2. \quad (2.6)$$

Combining the three terms in (2.4)–(2.6), the representation-learning objective is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{shared}} + \lambda_o \mathcal{L}_{\text{orth}} + \lambda_r \mathcal{L}_{\text{recon}}, \quad (2.7)$$

where $\lambda_o, \lambda_r > 0$ control the strengths of the last two terms. The shared loss places the outputs of the modality-specific shared encoders in a common coordinate system. The resulting shared and modality-specific representations determine which parts of a focal-pattern representation each observed pattern can supply.

2.3 Preliminary Prediction from Available Representations

For observations with pattern A_j , let $\mathbf{Z}_{(j)} = (\mathbf{z}_{1,(j)}, \dots, \mathbf{z}_{n_j,(j)})^\top \in \mathbb{R}^{n_j \times d_j}$ stack the representations indexed by S_j , and let $\mathbf{Y}_{(j)} \in \mathbb{R}^{n_j}$ be the corresponding response vector. MOSAIC

uses all patterns to construct a different preliminary predictor for each focal pattern. For focal pattern A_k and any observed pattern A_j , define $\mathbf{Z}_{(j \rightarrow k)} \in \mathbb{R}^{n_j \times d_k}$ by retaining the components indexed by $S_j \cap S_k$ and zero-filling those indexed by $S_k \setminus S_j$. Assuming that an observed modality-specific representation is nonzero almost surely, the zero-filled blocks distinguish unavailable representations from observed ones. This maps observations from every pattern into the focal-pattern input space $\mathbb{R}^{d_k}$, while retaining only the representation blocks supplied by their observed pattern.

To describe the focal-pattern information unavailable under A_j , define the complementary oracle matrix $\mathbf{Z}_{(-j \rightarrow k)} \in \mathbb{R}^{n_j \times d_k}$ by retaining the components indexed by $S_k \setminus S_j$ and zero-filling those indexed by $S_j \cap S_k$. Although these blocks are unobserved under A_j , they are well-defined in the latent representation model. Thus, $\mathbf{Z}_{(j \rightarrow k)}$ and $\mathbf{Z}_{(-j \rightarrow k)}$ partition the complete focal-pattern representation into blocks supplied and not supplied by A_j . For example, if $S_k = \{0, 2, 4\}$ and $S_j = \{0, 2\}$, then $\mathbf{z}_{i,(j \rightarrow k)} = (\mathbf{z}_i^{(0)\top}, \mathbf{z}_i^{(2)\top}, \mathbf{0}_{d(4)}^\top)^\top$ and $\mathbf{z}_{i,(-j \rightarrow k)} = (\mathbf{0}_{d(0)}^\top, \mathbf{0}_{d(2)}^\top, \mathbf{z}_i^{(4)\top})^\top$. Only $\mathbf{Z}_{(j \rightarrow k)}$ enters the estimator; the complementary matrix is used in the population analysis to characterize the effect of incomplete representation coverage. Let $\mathcal{H}_k$ be the first-stage function class. MOSAIC estimates the preliminary predictor by:

$$\hat{h}_k \in \arg \min_{h \in \mathcal{H}_k} \frac{1}{n} \sum_{j=1}^K \left\| \mathbf{Y}_{(j)} - h(\mathbf{Z}_{(j \rightarrow k)}) \right\|_2^2. \quad (2.8)$$

Let $\mathbf{z}_{(j \rightarrow k)}$ denote a generic row of $\mathbf{Z}_{(j \rightarrow k)}$. Conditioning on the realized pattern counts, the corresponding population preliminary predictor is:

$$h_k := \arg \min_{h \in \mathcal{H}_k} \sum_{j=1}^K \frac{n_j}{n} \mathbb{E} \left[\{Y - h(\mathbf{z}_{(j \rightarrow k)})\}^2 \mid \mathcal{A} = A_j \right].$$

2.4 Focal-Pattern Calibration

The population preliminary predictor h_k need not equal $g_{(k)}$ because other patterns can omit focal-pattern representations or have different conditional response relations. Calibration

estimates the resulting correction Δ_k from observations with pattern A_k , where every representation in S_k is available.

The statistical gain occurs when the representations required by $\hat{h}_k$ are supplied by enough observations across patterns and Δ_k is easier to estimate from the focal-pattern observations than the full rule. Let $\mathcal{D}_k$ be a class of calibration functions and let $P(\cdot)$ be a penalty. MOSAIC estimates the correction from focal-pattern observations:

$$\hat{\Delta}_k \in \arg \min_{\Delta \in \mathcal{D}_k} \frac{1}{n_k} \left\| \mathbf{Y}_{(k)} - \hat{h}_k(\mathbf{Z}_{(k)}) - \Delta(\mathbf{Z}_{(k)}) \right\|_2^2 + \lambda_k P(\Delta), \quad \hat{g}_{(k)} := \hat{h}_k + \hat{\Delta}_k. \quad (2.9)$$

Here, $P(\Delta)$ controls the correction complexity when n_k is small. Algorithm 1 summarizes the procedure.

Algorithm 1 MOSAIC estimation procedure

Require: unlabelled observations $\{(\mathcal{X}_i(\mathcal{A}_i), \mathcal{A}_i) : i = n + 1, \dots, n + N\}$ and labelled observations $\{(\mathcal{X}_i(\mathcal{A}_i), y_i, \mathcal{A}_i) : i = 1, \dots, n\}$; tuning parameters $\lambda_o, \lambda_r, \{\lambda_k\}_{k=1}^K$.

- 1: Estimate $\{\hat{U}^{(l)}\}_{l=1}^L$ and $\{\hat{V}^{(l)}\}_{l=1}^L$ by minimizing $\mathcal{L}_{\text{total}}$.
 - 2: **for** $i = 1, \dots, n$ **do**
 - 3: Compute $\hat{\mathbf{z}}_i^{(0,l)} = \hat{U}^{(l)}(\mathbf{x}_i^{(l)})$ and $\hat{\mathbf{z}}_i^{(l)} = \hat{V}^{(l)}(\mathbf{x}_i^{(l)})$ for every l with $m_i^{(l)} = 1$.
 - 4: Obtain the fused shared representation, for example by $\hat{\mathbf{z}}_i^{(0)} = \frac{\sum_{l=1}^L m_i^{(l)} \hat{\mathbf{z}}_i^{(0,l)}}{\sum_{l=1}^L m_i^{(l)}}$.
 - 5: Construct $\mathbf{Z}_{(k)}$ and $\mathbf{Y}_{(k)}$ for each focal pattern A_k .
 - 6: **for** $k = 1, \dots, K$ **do**
 - 7: Construct $\mathbf{Z}_{(j \rightarrow k)}$ for every $j = 1, \dots, K$.
 - 8: Estimate the preliminary predictor $\hat{h}_k$ by (2.8).
 - 9: Estimate the focal-pattern correction $\hat{\Delta}_k$ by (2.9).
 - 10: **return** $\{\hat{g}_{(k)} = \hat{h}_k + \hat{\Delta}_k : k = 1, \dots, K\}$.
-

2.5 Linear Model Specialization

We specialize the framework to linear prediction under the true complete-information model $Y = \boldsymbol{\beta}^{\star\top} \mathbf{z}^{\star} + \varepsilon$, with $\mathbb{E}(\varepsilon \mid \mathbf{z}^{\star}, \mathcal{A}) = 0$, where $\boldsymbol{\beta}^{\star} \in \mathbb{R}^d$ is the complete-information coefficient. For pattern A_k , partition $\mathbf{z}^{\star}$ into the available representation $\mathbf{z}_{(k)}^{\star}$ indexed by S_k and the complementary representation $\mathbf{z}_{(-k)}^{\star}$ indexed by $S_k^c = (\{0\} \cup [L]) \setminus S_k$. The linear specialization of (2.3) is:

$$\boldsymbol{\beta}_{(k)}^{\star} := \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^{d_k}} \mathbb{E} \left[\{Y - \boldsymbol{\beta}^{\top} \mathbf{z}_{(k)}^{\star}\}^2 \mid \mathcal{A} = A_k \right], \quad g_{(k)}(\mathbf{z}_{(k)}^{\star}) = \boldsymbol{\beta}_{(k)}^{\star\top} \mathbf{z}_{(k)}^{\star}.$$

Although $\boldsymbol{\beta}^{\star}$ is common, $\boldsymbol{\beta}_{(k)}^{\star}$ need not equal $\boldsymbol{\beta}_{S_k}^{\star}$ because the unavailable representation $\mathbf{z}_{(-k)}^{\star}$ may be associated with $\mathbf{z}_{(k)}^{\star}$. The following scalar two-pattern example shows how an unavailable but correlated representation changes a pattern-specific coefficient and why a pooled model that imposes a common coefficient on $z^{(1)}$ cannot generally recover both pattern-specific rules.

Proposition 2 (Mismatch under indicator-augmented pooling) *Let $d^{(0)} = d^{(1)} = d^{(2)} = 1$, write $\boldsymbol{\beta}^{\star} = (\beta_0^{\star}, \beta_1^{\star}, \beta_2^{\star})^{\top}$, and suppose that $z^{\star,(0)}$, $z^{\star,(1)}$, and $z^{\star,(2)}$ are mean-zero and unit-variance, with $z^{\star,(0)} \perp (z^{\star,(1)}, z^{\star,(2)})$ and $\text{Cov}(z^{\star,(1)}, z^{\star,(2)}) = \rho$. Consider two observed-modality patterns $A_1 = \{1\}$ and $A_2 = \{1, 2\}$, with representation index sets $S_1 = \{0, 1\}$ and $S_2 = \{0, 1, 2\}$. Let $m^{(2)} = \mathbf{1}\{\mathcal{A} = A_2\} \sim \text{Bernoulli}(\pi)$ be independent of $(z^{\star,(0)}, z^{\star,(1)}, z^{\star,(2)}, \varepsilon)$. Then the pattern-specific linear coefficients are $\boldsymbol{\beta}_{(1)}^{\star} = (\beta_0^{\star}, \beta_1^{\star} + \rho\beta_2^{\star})^{\top}$ and $\boldsymbol{\beta}_{(2)}^{\star} = (\beta_0^{\star}, \beta_1^{\star}, \beta_2^{\star})^{\top}$, whereas the population coefficient from the pooled design $(z^{\star,(0)}, z^{\star,(1)}, m^{(2)}, m^{(2)}z^{\star,(2)})^{\top}$ is*

$$\boldsymbol{\beta}_{\text{pool}}^{\star} = \left(\beta_0^{\star}, \beta_1^{\star} + \frac{(1 - \pi)\rho\beta_2^{\star}}{1 - \pi\rho^2}, 0, \frac{(1 - \rho^2)\beta_2^{\star}}{1 - \pi\rho^2} \right)^{\top}.$$

Except in degenerate cases such as $\rho\beta_2^{\star} = 0$, the rules induced by $\boldsymbol{\beta}_{\text{pool}}^{\star}$ do not coincide with either pattern-specific rule.

The pooled model in Proposition 2 constrains the coefficient of $z^{\star,(1)}$ to be the same in both patterns. This restriction can be relaxed by allowing covariate coefficients to depend on the missingness pattern, for example through covariate–indicator interactions (Fletcher Mercaldo & Blume 2020, Bertsimas et al. 2025). In the present example, adding $m^{(2)}z^{\star,(1)}$ recovers both pattern-specific rules at the population level. However, increasing such pattern indicator introduces additional pattern-dependent parameters. It reduces to separate pattern-specific fitting (Fletcher Mercaldo & Blume 2020, Bertsimas et al. 2025), which can be inefficient when some pattern samples are small. This motivates borrowing across patterns while retaining target-pattern calibration rather than imposing either common or fully pattern-specific coefficients.

We now return to the general K -pattern model and specialize the two stages of MOSAIC. For a focal pattern A_k and every observed pattern A_j , write in matrix form:

$$\mathbf{Y}_{(j)} = \left(\mathbf{Z}_{(j \rightarrow k)}^* + \mathbf{Z}_{(-j \rightarrow k)}^* \right) \boldsymbol{\beta}_{(k)}^* + \mathbf{r}_{(j,k)}^*, \quad j = 1, \dots, K,$$

where $\mathbf{r}_{(j,k)}^*$ is the residual from applying the focal-pattern rule to observations with pattern A_j . If $\mathcal{H}_k = \{h_{\mathbf{w}} : h_{\mathbf{w}}(\mathbf{Z}) = \mathbf{Z}\mathbf{w}, \mathbf{w} \in \mathbb{R}^{d_k}\}$, then (2.8) becomes:

$$\widehat{\mathbf{w}}_{(k)}^* \in \arg \min_{\mathbf{w} \in \mathbb{R}^{d_k}} \frac{1}{n} \sum_{j=1}^K \left\| \mathbf{Y}_{(j)} - \mathbf{Z}_{(j \rightarrow k)}^* \mathbf{w} \right\|_2^2, \quad \widehat{h}_k(\mathbf{Z}) = \mathbf{Z} \widehat{\mathbf{w}}_{(k)}^*.$$

Its population coefficient is $\mathbf{w}_{(k)}^* := \arg \min_{\mathbf{w} \in \mathbb{R}^{d_k}} \sum_{j=1}^K \frac{n_j}{n} \mathbb{E} \left[\{Y - \mathbf{z}_{(j \rightarrow k)}^{\star, \top} \mathbf{w}\}^2 \mid \mathcal{A} = A_j \right]$. Define the corresponding weighted moment matrices as:

$$\widetilde{\boldsymbol{\Sigma}}_{(k)}^{oo,*} = \sum_{j=1}^K \frac{n_j}{n} \widetilde{\boldsymbol{\Sigma}}_{j,k}^{oo,*}, \quad \widetilde{\boldsymbol{\Sigma}}_{(k)}^{om,*} = \sum_{j=1}^K \frac{n_j}{n} \widetilde{\boldsymbol{\Sigma}}_{j,k}^{om,*},$$

where $\widetilde{\boldsymbol{\Sigma}}_{j,k}^{oo,*} = \mathbb{E}[\mathbf{z}_{(j \rightarrow k)}^* \mathbf{z}_{(j \rightarrow k)}^{\star, \top} \mid \mathcal{A} = A_j]$, $\widetilde{\boldsymbol{\Sigma}}_{j,k}^{om,*} = \mathbb{E}[\mathbf{z}_{(j \rightarrow k)}^* \mathbf{z}_{(-j \rightarrow k)}^{\star, \top} \mid \mathcal{A} = A_j]$. Under the cross-pattern moment condition $\mathbb{E}[\mathbf{z}_{(j \rightarrow k)}^* r_{(j,k)}^* \mid \mathcal{A} = A_j] = \mathbf{0}$ for every $j \in [K]$, the normal equations give $\mathbf{w}_{(k)}^* = \boldsymbol{\beta}_{(k)}^* + \left(\widetilde{\boldsymbol{\Sigma}}_{(k)}^{oo,*} \right)^{-1} \widetilde{\boldsymbol{\Sigma}}_{(k)}^{om,*} \boldsymbol{\beta}_{(k)}^*$. Thus, the coefficient of the preliminary predictor can differ from $\boldsymbol{\beta}_{(k)}^*$. Write the coefficient correction as $\boldsymbol{\delta}_{(k)}^* := \boldsymbol{\beta}_{(k)}^* - \mathbf{w}_{(k)}^*$ and

define $\tilde{\rho}_{(k)}^* = \|(\tilde{\Sigma}_{(k)}^{oo,*})^{-1} \tilde{\Sigma}_{(k)}^{om,*}\|_{\text{op}}$. This quantity measures how strongly the representations supplied across patterns are associated with focal-pattern representations omitted in some patterns, and $\|\delta_{(k)}^*\|_2 \leq \tilde{\rho}_{(k)}^* \|\beta_{(k)}^*\|_2$. If $\mathcal{D}_k = \{\Delta_{\delta} : \Delta_{\delta}(\mathbf{Z}) = \mathbf{Z}\delta, \delta \in \mathbb{R}^{d_k}\}$ and $P(\Delta) = \|\delta\|_2$, the calibration estimator becomes:

$$\hat{\delta}_{(k)}^* \in \arg \min_{\delta \in \mathbb{R}^{d_k}} \frac{1}{n_k} \left\| \mathbf{Y}_{(k)} - \mathbf{Z}_{(k)}^* (\widehat{\mathbf{w}}_{(k)}^* + \delta) \right\|_2^2 + \lambda_k \|\delta\|_2, \quad \hat{\beta}_{(k)}^* := \widehat{\mathbf{w}}_{(k)}^* + \hat{\delta}_{(k)}^*.$$

3 Theory

The theory below specializes MOSAIC to the linear model in Section 2.5. We first study cross-pattern borrowing and focal-pattern calibration under oracle representations. We then establish recovery of the representations and finally propagate representation error into prediction. We first distinguish the true representations $\mathbf{z}_i^{*,(l)}$ from their learned counterparts $\hat{\mathbf{z}}_i^{(l)}$ and then introduce the assumptions required for the theoretical analysis.

Assumption 1 (Missing Completely at Random (MCAR)) *Assume that $\mathbf{m}_i = (m_i^{(1)}, \dots, m_i^{(L)})^\top$ is independent of $(\mathbf{z}_i^*, \varepsilon_{x,i}, Y_i)$. For every $l, l' \in [L]$ and missingness pattern $k \in [K]$, $\Pr(m_i^{(l)} = 1, m_i^{(l')} = 1) = q^{(ll')} \geq \underline{q}_N > 0$ and $\pi_k \geq \underline{q}_{n,k} > 0$, where $\underline{q}_N \asymp N^{-\gamma}$, $\underline{q}_{n,k} \asymp n^{-\iota_k}$, $\gamma, \iota_k \in [0, 1)$. Let $\mathcal{H}^s = \{k \in [K] : |A_k| = 1\}$. We further assume that there exists a constant $c_\iota > 0$ such that $\min_{k \in [K] \setminus \mathcal{H}^s} \iota_k - 2 \max_{k \in \mathcal{H}^s} \iota_k \geq c_\iota$.*

Assumption 2 (Prediction Noise) *For each $k, j \in [K]$: (i) $\mathbb{E}[Y - \beta_{(k)}^{*\top} \mathbf{z}_{(k)}^* \mid \mathcal{A} = A_k] = 0$; (ii) $\mathbb{E}\{\mathbf{z}_{(j \rightarrow k),i}^* t^*_{(j,k),i} \mid \mathcal{A} = A_j\} = \mathbf{0}$; (iii) For any $t \in \mathbb{R}$, $\mathbb{E}\{\exp(tr_{(j,k),i}^*) \mid \mathcal{A} = A_j\} \leq \exp(\frac{1}{2} \tau^2 t^2)$.*

Assumption 3 (Representation) *For each representation $l \in \{0\} \cup [L]$: (i) $\mathbb{E}[\mathbf{z}_i^{*,(l)}] = \mathbf{0}$; (ii) $\text{Cov}(\mathbf{z}_i^{*,(l)}) = \mathbf{I}_{d(l)}$; (iii) For any unit $\mathbf{u} \in \mathbb{R}^{d(l)}$ and $t \in \mathbb{R}$, $\mathbb{E}[\exp\{t \langle \mathbf{u}, \mathbf{z}_i^{*,(l)} \rangle\}] \leq \exp(\frac{1}{2} \tau^2 t^2)$; (iv) $\text{Cov}(\mathbf{z}_i^{*,(l)}, \mathbf{z}_i^{*,(l')}) = \mathbf{0}$ for $l \neq l'$.*

For the theoretical analysis, Assumption 1 allows the relevant observation and pattern

probabilities to decay with sample size under MCAR controlled by γ and ι_k . The final condition imposes a separation between the rarity orders of single-modality and multiple-modalities patterns; its implication for cross-pattern borrowing is discussed after Theorem 1. Assumption 2 provides the moment and tail conditions for the noise, and Assumption 3 controls the covariance structure and sub-Gaussian tails of the true representations. The unit-covariance normalization fixes their scale.

Let $n_{k,(l)} = \sum_{j: l \in S_j} n_j$ be the number of first-stage observations supplying focal-pattern representation $l \in S_k$. Define the effective sample size as $\tilde{n}_{(k)} := n \left\{ \frac{\min_{l \in S_k} n_{k,(l)}}{n} \right\}^2$. We are now ready to establish the oracle prediction error bound.

Theorem 1 *Fix a focal pattern A_k and condition on the realized missingness patterns. Suppose that $n_k \gtrsim d_k + \log n$ and $\tilde{n}_{(k)} \gtrsim d_k + \log n$. Under Assumptions 1, 2, and 3(i)–(iii), if $\lambda_k \asymp \sqrt{(d_k + \log n)/n_k}$, then, with probability at least $1 - Cn^{-1}$,*

$$\|\hat{\boldsymbol{\beta}}_{(k)}^* - \boldsymbol{\beta}_{(k)}^*\|_2^2 \lesssim \left\{ 1 + (1 + \tilde{\rho}_{(k)}) \|\boldsymbol{\beta}_{(k)}^*\|_2 \right\}^2 \frac{d_k + \log n}{\tilde{n}_{(k)}} + \min \left\{ \frac{d_k + \log n}{n_k}, \|\boldsymbol{\delta}_{(k)}^*\|_2^2 \right\}. \quad (3.1)$$

If Assumption 3(iv) also holds, $\|\boldsymbol{\beta}_{(k)}^\|_2$ is uniformly bounded, and the multiplicative constant in λ_k is sufficiently large, then simultaneously for all $l \in S_k$,*

$$\|\hat{\boldsymbol{\beta}}_{(k)}^{*,(l)} - \boldsymbol{\beta}_{(k)}^{*,(l)}\|_2^2 \lesssim \left(1 + \|\boldsymbol{\beta}_{(k)}^*\|_2 \right)^2 \frac{d_k + \log n}{n_{k,(l)}}. \quad (3.2)$$

In particular, since $n_{k,(0)} = n$, $\|\hat{\boldsymbol{\beta}}_{(k)}^{,(0)} - \boldsymbol{\beta}_{(k)}^{*,(0)}\|_2^2 \lesssim \left(1 + \|\boldsymbol{\beta}_{(k)}^*\|_2 \right)^2 \frac{d_k + \log n}{n}$.*

The first term in (3.1) is the preliminary-predictor estimation error and is governed by the focal-pattern representation supplied by the fewest observations. The second term is the calibration error: a small correction is inexpensive, while a large correction returns this component to the local scale. Thus, the general bound makes explicit how representation coverage and focal-pattern correction jointly determine the gain from cross-pattern borrowing. Under Assumption 3(iv), the population correction vanishes, and a sufficiently large calibration penalty also removes the empirical correction. The componentwise bound in

(3.2) then retains the observation count for each representation block rather than reducing all blocks to the least-covered representation.

For the shared representation, $n_{k,(0)} = n$. Thus, when $n_k = o(n)$, the shared coefficient has a strictly faster rate under cross-pattern borrowing. This all-pattern gain can include observations with no raw covariate in common with the focal pattern: if $A_j \cap A_k = \emptyset$, then $S_j \cap S_k = \{0\}$, so the n_j observations with pattern A_j contribute to $n_{k,(0)}$. For comparison, we modify the linear model in Section 2.5 by replacing the latent representations with the corresponding raw modalities. For each raw modality l , we decompose its coefficient under missingness pattern k as a common coefficient vector $\boldsymbol{\theta}^{*,(l)}$ plus a sparse pattern-specific deviation $\boldsymbol{\Delta}_k^{*,(l)}$, for $l \in A_k$. This formulation also covers SPSM (Stempfle et al. 2023). Proposition 3 then provides a theoretical result under this raw-modality formulation.

Proposition 3 (Model without representations) *Fix a focal pattern A_k . For each $l \in A_k$, write $\boldsymbol{\theta}^{*,(l)} + \boldsymbol{\Delta}_k^{*,(l)}$ for the coefficient of raw modality l under A_k , and let $s_k^{(l)} = \|\boldsymbol{\Delta}_k^{*,(l)}\|_0$. Suppose that the model is correctly specified. Under the regularity conditions stated in Supplementary Section A.5, with probability at least $1 - Cn^{-1}$, simultaneously for all $l \in A_k$,*

$$\left\| (\widehat{\boldsymbol{\theta}}^{(l)} + \widehat{\boldsymbol{\Delta}}_k^{(l)}) - (\boldsymbol{\theta}^{*,(l)} + \boldsymbol{\Delta}_k^{*,(l)}) \right\|_2^2 \lesssim \frac{d_{\text{raw}}^{(l)} + \log n}{n_{k,(l)}} + \min \left\{ \frac{s_k^{(l)} \log \{nd_{\text{raw}}^{(l)}\}}{n_k}, \|\boldsymbol{\Delta}_k^{*,(l)}\|_2^2 \right\}. \quad (3.3)$$

Proposition 3 and Theorem 1 show two further gains from MOSAIC, beyond $n_{k,(0)} = n$. First, representation replaces the raw covariate dimension by the much smaller focal-pattern representation dimension d_k , yielding a lower-dimensional estimation problem whenever $d_k \ll d_{\text{raw}}^{(l)}$. Second, if we do not disentangle the raw covariates into shared and specific representations, orthogonality between the raw covariates is not guaranteed. In contrast, we impose Assumption 3(iv), allowing the second term in the error bound to vanish.

In addition, the separation condition in Assumption 1 yields a strict improvement over focal-only fitting, by describing a setting in which patterns containing multiple modalities

are less frequent than single-modality patterns. If every $l \in A_k$ has a corresponding single-modality pattern $h^{(l)}$, then $n_{k,(l)} \geq n_{h^{(l)}}$ and, with high probability, $n_{k,(l)} \gtrsim n^{1-\max_{j \in \mathcal{H}^s} \iota_j}$ for every $l \in A_k$. If the lower envelope is also sharp for a multi-modality focal pattern, so that $\pi_k \asymp n^{-\iota_k}$ and hence $n_k \asymp n^{1-\iota_k}$ with high probability, then $n_{k,(l)}/n_k \gtrsim n^{\iota_k - \max_{j \in \mathcal{H}^s} \iota_j} \gtrsim n^{c_\iota} \rightarrow \infty$. Therefore, under (3.2), every focal modality-specific coefficient block has an error of $o\{(d_k + \log n)/n_k\}$, and the shared coefficient has the all-pattern rate based on n . In the general case without Assumption 3(iv), the same count comparison gives $\tilde{n}_{(k)} \gtrsim n^{1-2\max_{j \in \mathcal{H}^s} \iota_j}$ and $\tilde{n}_{(k)}/n_k \gtrsim n^{c_\iota} \rightarrow \infty$. When $\{1 + (1 + \tilde{\rho}_{(k)}^*)\|\boldsymbol{\beta}_{(k)}^*\|_2\}^2 = O(1)$ and $\|\boldsymbol{\delta}_{(k)}^*\|_2^2 = o\{(d_k + \log n)/n_k\}$, (3.1) then gives an overall error of $o\{(d_k + \log n)/n_k\}$.

Theorem 1 is conditional on the realized pattern counts. Supplementary Section B.2 shows that replacing the realized proportions n_j/n by the population prevalences π_j introduces only a perturbation; for fixed K , the additional squared-error term is of order $\|\boldsymbol{\beta}_{(k)}^*\|_2^2 \log n/n$.

We next study recovery of the oracle representations used in Theorem 1. For notational simplicity, we re-index the unlabelled sample as $i = 1, \dots, N$; the independent labelled sample continues to have size n . We define $\mathbf{x}_i^{(l)} \in \mathbb{R}^{d_{\text{raw}}^{(l)}}$ using (2.2) with fixed matrices $\mathbf{A}^{(l)} \in \mathbb{R}^{d_{\text{raw}}^{(l)} \times d^{(0)}}$, $\mathbf{B}^{(l)} \in \mathbb{R}^{d_{\text{raw}}^{(l)} \times d^{(l)}}$, and $\mathbf{C}^{(l)} \in \mathbb{R}^{d_{\text{raw}}^{(l)} \times d_{\text{noise}}^{(l)}}$:

$$\mathbf{x}_i^{(l)} = \mathbf{A}^{(l)} \mathbf{z}_i^{\star, (0)} + \mathbf{B}^{(l)} \mathbf{z}_i^{\star, (l)} + \mathbf{C}^{(l)} \boldsymbol{\epsilon}_{x,i}^{(l)}, \quad (3.4)$$

where $d_{\text{noise}}^{(l)} := d_{\text{raw}}^{(l)} - d^{(0)} - d^{(l)}$ and $\max(d^{(0)}, d^{(l)}) \ll d_{\text{noise}}^{(l)}$. This high-dimensional ambient-noise regime is used only to analyse representation recovery; it is not required by Algorithm 1.

We model the corresponding encoders as matrices $\mathbf{U}^{(l)}$ and $\mathbf{V}^{(l)}$. The ideal encoders that recover the shared and modality-specific representations separately are $\mathbf{U}_{\text{ideal}}^{(l)} = (\mathbf{A}^{(l)\top} \mathbf{A}^{(l)})^{-1} \mathbf{A}^{(l)\top}$ and $\mathbf{V}_{\text{ideal}}^{(l)} = (\mathbf{B}^{(l)\top} \mathbf{B}^{(l)})^{-1} \mathbf{B}^{(l)\top}$.

Let the joint encoder be defined as $\mathbf{W}^{(l)} = [\mathbf{U}^{(l)\top}, \mathbf{V}^{(l)\top}]^\top \in \mathbb{R}^{(d^{(0)}+d^{(l)}) \times d_{\text{raw}}^{(l)}}$, and let $\mathbf{T}^{(l)}$ be the corresponding linear decoder matrix. Define the number of co-observed samples for modality pair (l, k) as $N_{lk} = \sum_{i=1}^N m_i^{(l)} m_i^{(k)}$ and the aggregated empirical covariance of

modality l as $\tilde{\Sigma}_{ll} = L^{-1} \sum_{k=1}^L N_{lk}^{-1} \sum_{i=1}^N m_i^{(l)} m_i^{(k)} \mathbf{x}_i^{(l)} \mathbf{x}_i^{(k)\top}$. We work on the event that all N_{lk} are positive; Assumption 1 and the stated sample-size conditions ensure that this event has high probability. For a tractable linear analysis, we study the following surrogate of (2.7):

$$\begin{aligned} \min_{\{\mathbf{W}^{(l)}, \mathbf{T}^{(l)}\}_{l=1}^L} & \frac{1}{L^2} \sum_{l,k=1}^L \frac{1}{N_{lk}} \sum_{i=1}^N m_i^{(l)} m_i^{(k)} \left(\|\mathbf{U}^{(l)} \mathbf{x}_i^{(l)} - \mathbf{U}^{(k)} \mathbf{x}_i^{(k)}\|_2^2 + \lambda \|\mathbf{T}^{(l)} \mathbf{W}^{(l)} \mathbf{x}_i^{(l)} - \mathbf{x}_i^{(l)}\|_2^2 \right) \\ \text{s.t. } & \mathbf{W}^{(l)} \tilde{\Sigma}_{ll} \mathbf{W}^{(l)\top} = \mathbf{I}_{d^{(0)}+d^{(l)}}, \end{aligned}$$

where $\lambda > 0$ is a tuning parameter. This surrogate replaces the soft orthogonalization penalty in (2.7) by a hard whitening constraint and uses a single weight for reconstruction. The hard constraint can be imposed directly in the linear model; details are provided in Supplementary Section A.2. We further impose the following assumptions.

Assumption 4 (Structure) *For every modality l , $\mathbf{A}^{(l)}$ and $\mathbf{B}^{(l)}$ have full column rank, while $\mathbf{C}^{(l)}$ is allowed to be rank-deficient. Furthermore, $(\mathbf{A}^{(l)})^\top \mathbf{B}^{(l)} = \mathbf{0}$, $(\mathbf{A}^{(l)})^\top \mathbf{C}^{(l)} = \mathbf{0}$, and $(\mathbf{B}^{(l)})^\top \mathbf{C}^{(l)} = \mathbf{0}$. Let $\sigma_i(\mathbf{M})$ denote the i -th singular value of a matrix $\mathbf{M}$. There exist positive constants $0 < \nu \leq \mu$ such that every singular value of $\mathbf{A}^{(l)}$ and $\mathbf{B}^{(l)}$ satisfies $\sigma_i(\mathbf{A}^{(l)}), \sigma_i(\mathbf{B}^{(l)}) \in [\nu, \mu]$, and every singular value of $\mathbf{C}^{(l)}$ satisfies $\sigma_i(\mathbf{C}^{(l)}) \leq \frac{1}{\sqrt{2}}\nu$.*

Assumption 5 (Representation Noise) *For each modality $l \in [L]$: (i) $\mathbb{E}[\boldsymbol{\varepsilon}_{x,i}^{(l)}] = \mathbf{0}$; (ii) $\text{Cov}(\boldsymbol{\varepsilon}_{x,i}^{(l)}, \mathbf{z}_i^{*,(k)}) = \mathbf{0}$ for $k \in \{0, \dots, L\}$; (iii) $\text{Cov}(\boldsymbol{\varepsilon}_{x,i}^{(l)}) = \mathbf{I}_{d_{\text{noise}}^{(l)}}$; (iv) For any unit $\mathbf{u} \in \mathbb{R}^{d_{\text{noise}}^{(l)}}$ and $t \in \mathbb{R}$, $\mathbb{E}[\exp\{t\langle \mathbf{u}, \boldsymbol{\varepsilon}_{x,i}^{(l)} \rangle\}] \leq \exp(\frac{1}{2}\tau^2 t^2)$; (v) $\text{Cov}(\boldsymbol{\varepsilon}_{x,i}^{(l)}, \boldsymbol{\varepsilon}_{x,i}^{(k)}) = \mathbf{0}$ for $l \neq k$.*

Assumption 4 imposes geometric separation of the shared and modality-specific representation subspaces, which makes the representations identifiable up to an orthogonal transformation in the linear analysis. Assumption 5 controls the sub-Gaussian tails of the modality-level noise. Under Assumptions 3–5, the shared representation is identifiable up to one common orthogonal transformation across modalities, while each modality-specific representation is identifiable up to its own orthogonal transformation; the proof is given in Supplementary Section A.2. Let $d_{\max} = \max_{l \in [L]} d_{\text{raw}}^{(l)}$ and let $\mathcal{O}(r)$ denote the set of $r \times r$

orthogonal matrices. The following theorem bounds the distances between the estimated encoders and their ideal counterparts, up to these transformations.

Theorem 2 *Under Assumption 1 and Assumptions 3–5, suppose the sample size N is sufficiently large such that $\epsilon_{N,d_{\max}} \leq \frac{1}{8}\nu^2$, and the regularization parameter satisfies $\lambda \leq \min(\epsilon_{N,d_{\max}}, \frac{L-2}{L\mu^2})$ with $L \geq 3$, where $\epsilon_{N,d_{\max}} = C_1(d_{\max} + \log N)^{1/2}N^{-(1-\gamma)/2}$ is detailed in the proof. Then, with probability at least $1 - c \exp(-c'N^{1-\gamma}) - CN^{-1}$, the empirical encoders $(\widehat{\mathbf{U}}^{(l)}, \widehat{\mathbf{V}}^{(l)})$ converge to $(\mathbf{U}_{\text{ideal}}^{(l)}, \mathbf{V}_{\text{ideal}}^{(l)})$ up to the corresponding orthogonal transformations at the following rate:*

$$\begin{aligned} \max \left\{ \min_{\mathbf{R}_0 \in \mathcal{O}(d^{(0)})} \frac{1}{L} \sum_{l=1}^L \|\widehat{\mathbf{U}}^{(l)} - \mathbf{R}_0 \mathbf{U}_{\text{ideal}}^{(l)}\|_F, \frac{1}{L} \sum_{l=1}^L \min_{\mathbf{R}_l \in \mathcal{O}(d^{(l)})} \|\widehat{\mathbf{V}}^{(l)} - \mathbf{R}_l \mathbf{V}_{\text{ideal}}^{(l)}\|_F \right\} \\ \lesssim \{d_{\max}(d_{\max} + \log N)\}^{1/2} N^{-(1-\gamma)/2}, \end{aligned}$$

where C, c, c' are positive constants independent of $(N, d_{\max})$.

Theorem 2 establishes that the shared and modality-specific representations are recoverable. Since $\gamma < 1$, the representation error converges to zero as the unlabelled sample size N increases. The restriction $L \geq 3$ is used for spectral separation in this theorem; Algorithm 1 also applies to two modalities. We use the bound below to propagate representation-learning error into prediction. For a labelled subject, define $\widehat{\mathbf{z}}_i^{(0)} = \{\sum_{l=1}^L m_i^{(l)}\}^{-1} \sum_{l=1}^L m_i^{(l)} \widehat{\mathbf{U}}^{(l)} \mathbf{x}_i^{(l)}$ and $\widehat{\mathbf{z}}_i^{(l)} = \widehat{\mathbf{V}}^{(l)} \mathbf{x}_i^{(l)}$. The common unidentified rotation of the shared representation and the separate rotations of the modality-specific representations are absorbed into the corresponding oracle coordinate systems; rotating the associated coefficients in the same way leaves the prediction rule unchanged.

We next consider the learned-representation estimator used by MOSAIC. Under the orthogonality conditions in Assumption 4, the ideal encoders recover the true representations exactly, satisfying $\mathbf{U}_{\text{ideal}}^{(l)} \mathbf{x}_i^{(l)} = \mathbf{z}_i^{\star, (0)}$ and $\mathbf{V}_{\text{ideal}}^{(l)} \mathbf{x}_i^{(l)} = \mathbf{z}_i^{\star, (l)}$. Together with Theorem 2, this means that the learned design matrices can be treated as perturbations of the oracle design matrices in Theorem 1. Assumption 3 makes the population correction vanish in this

orthogonal linear specialization, so the next theorem isolates the additional cost of learning the representations.

Theorem 3 *Suppose the conditions of Theorem 1 hold, except that the tuning parameter is specified below, and let $\hat{\beta}_{(k)}$ be the estimator constructed from the learned representations $\hat{\mathbf{z}}_i^{(l)}$. Let $\eta_N = \{d_{\max}(d_{\max} + \log N)\}^{1/2} N^{-(1-\gamma)/2}$, and assume that the representation-learning and labelled samples are independent. Define $\bar{\iota}_k = \max_{l \in S_k} \min_{j: l \in S_j} \iota_j$, and suppose that the realized pattern counts satisfy $\max_{l \in S_k} n_{k,(l)} / \min_{l \in S_k} n_{k,(l)} \lesssim n^{\bar{\iota}_k}$. Suppose that $n^{\bar{\iota}_k} \eta_N + \sqrt{(d_k + \log N)/n} + \sqrt{(d_k + \log N)/n_k}$ is sufficiently small and that $\lambda_k \asymp \sqrt{(d_k + \log N)/n_k} + \eta_N$. Under Assumptions 1–5, with probability at least $1 - c \exp(-c' N^{1-\gamma}) - CN^{-1}$,*

$$\|\hat{\beta}_{(k)} - \beta_{(k)}^*\|_2^2 \lesssim \left(1 + \|\beta_{(k)}^*\|_2\right)^2 \left\{ \frac{d_k + \log N}{\tilde{n}_{(k)}} + n^{2\bar{\iota}_k} \eta_N^2 \right\}. \quad (3.5)$$

The first term in Theorem 3 is the oracle prediction rate. The second is the representation-learning cost, amplified when the numbers of observations supplying different focal-pattern representations are imbalanced. When the second term is smaller, the learned-representation estimator has the same leading rate as its oracle counterpart. Supplementary Section A.1 extends this result to generalized linear models, allowing representation-learning error and a potentially nonzero focal-pattern correction to enter the same bound; proofs are provided in Supplementary Appendix D.

4 Simulation Studies

The simulations are organized around the two stages of MOSAIC. Experiment 1 assesses recovery of the shared and modality-specific representations under missing modalities. Experiment 2 assesses downstream prediction after representation learning, with emphasis on whether overlap borrowing followed by focal-pattern correction improves performance across missingness patterns.

4.1 Experiment 1: Representation Recovery

Experiment 1 examines the representation-learning stage by varying the sample size N and the modality observation probability q . We also include ablations that remove orthogonal decoupling, reconstruction, or both, so that the roles of the representations can be separated. We generate $L = 3$ modalities from a latent structure with shared and modality-specific representations. Each subject has a shared representation $\mathbf{z}_i^{*,(0)} \in \mathbb{R}^5$, modality-specific representations $\mathbf{z}_i^{*,(l)} \in \mathbb{R}^5$, and noise. We consider a linear anisotropic mixing regime and a nonlinear residual-tanh regime, with details given in Supplementary Section E.1. For each subject and modality, $m_i^{(l)} \sim \text{Bernoulli}(q)$ independently.

We compare MOSAIC with three ablations: MOSAIC (w/o orth), which removes the covariance-orthogonality penalty; MOSAIC (w/o recon), which removes reconstruction; and MOSAIC (w/o orth & recon), which keeps only the shared alignment loss. We also include spectral SVD, a non-iterative baseline based on pairwise cross-modal covariance and residual within-modality variation, and the complete algorithm is given in Supplementary Section E.1. For the sample-size sweep, we fix $q = 0.6$ and vary N . For the missingness sweep, we fix $N = 5000$ and vary q . Each condition uses 100 Monte Carlo replications. Recovery is measured by no-scale Procrustes error at the subject-embedding level after column centring, with details given in Supplementary Section E.1.

Figure 3 shows that representation recovery generally improves as N increases, with the nonlinear observation setting being substantially harder than the linear setting. The main separation among methods appears in modality-specific recovery. Without orthogonal decoupling, the specific representation tends to retain variation that should be attributed to the shared representation; without reconstruction, the learned representation is less effective at preserving modality-specific information after alignment. Overall, MOSAIC is the most stable across the two recovery targets.

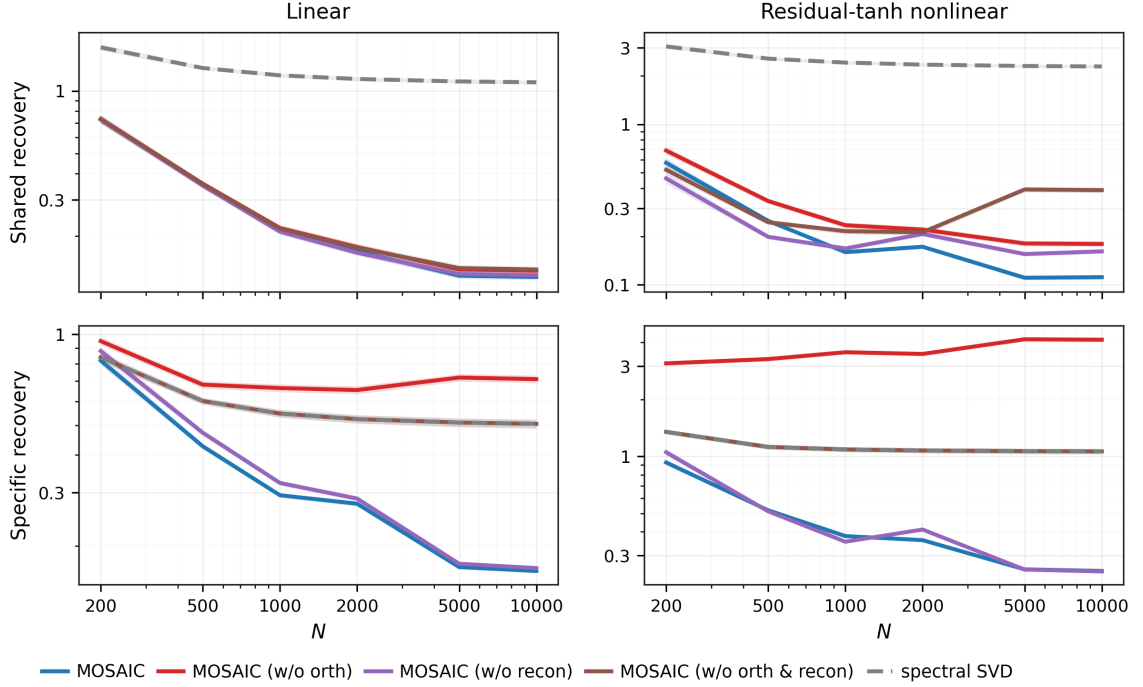


Figure 3: Representation recovery in the Experiment 1 sample-size sweep. Shaded bands are pointwise 95% Monte Carlo confidence intervals for the mean over 100 replications.

Figure 4 shows a complementary pattern as the modality observation probability q varies. Larger q improves recovery because more subjects have co-observed modalities, which strengthens the empirical alignment signal. This benefit is clearest for methods that can use alignment without collapsing the representation structure. The spectral SVD baseline is limited by its linear covariance construction, while the ablations remain more sensitive to leakage between shared and modality-specific components. Optimization and tuning details are provided in Supplementary Section E.1.

4.2 Experiment 2: Prediction under Missingness Patterns

Experiment 2 evaluates downstream prediction after representation learning. We first compare MOSAIC with missing-modality learning baselines under the same shared and modality-specific latent structure as Experiment 1. We then compare MOSAIC with the

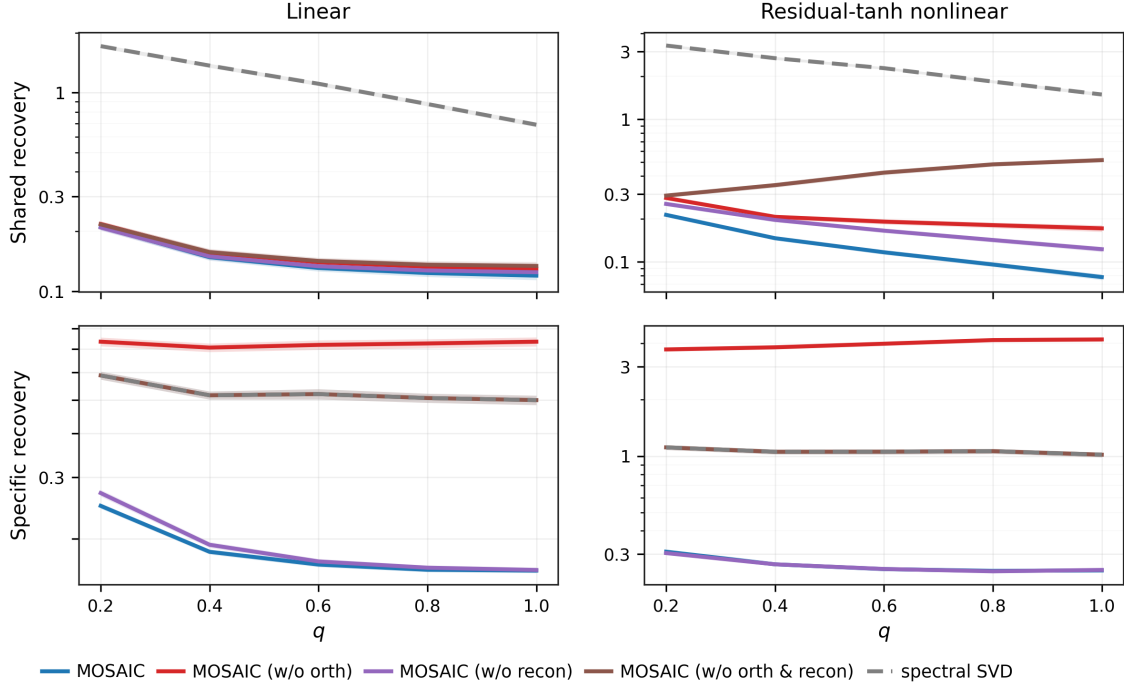


Figure 4: Representation recovery as q varies, with $N = 5000$. Shaded bands are pointwise 95% Monte Carlo confidence intervals for the mean over 100 replications.

sharing pattern submodels (SPSM) estimator of [Stempfle et al. \(2023\)](#), applied to the same aligned representations used by MOSAIC; we refer to this baseline as Aligned-SPSM. We also use four internal variants to assess overlap borrowing, focal-pattern calibration, pooled fitting, and representation learning. We maintain $L = 3$ modalities, and the covariate generation settings for the linear and nonlinear regimes follow Experiment 1. Responses are generated from the complete latent representation with additive Gaussian noise; detailed coefficient generation and training settings are in Supplementary Section E.2.

The external baselines are M3Care ([Zhang et al. 2022](#)), MedFuse ([Hayat et al. 2022](#)), ShaSpec ([Wang et al. 2023](#)), IF-MMIN ([Zhao et al. 2021](#), [Zuo et al. 2023](#)), TFR-Net ([Yuan et al. 2021](#)), and CorrKD ([Li et al. 2024](#)). Brief descriptions of the comparison methods are provided in Supplementary Section E.11. All methods use the same missingness indicators $m_i^{(l)}$, train-validation split, and early-stopping rule within each replication. We fix $N = 5000$

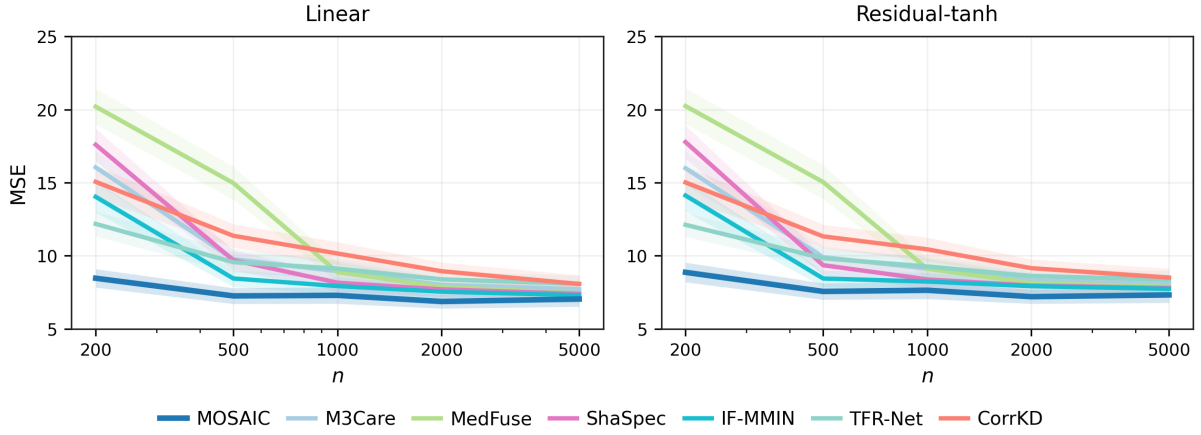


Figure 5: Comparison between MOSAIC and external baselines in Experiment 2. Shaded bands are pointwise 95% Monte Carlo confidence intervals for the mean over 100 replications.

and vary n . For evaluation, we generate a balanced test set with 200 subjects from each nonempty missingness pattern. We first compute the test MSE within each pattern, denoted by MSE_k , and report the equally weighted average $\text{MSE}_{\text{bal}} = K^{-1} \sum_{k=1}^K \text{MSE}_k$ to prevent common patterns from dominating the metric; with $L = 3$, this gives $K = 7$ patterns.

Figure 5 shows that MOSAIC achieves the lowest balanced-pattern MSE in both observation settings. IF-MMIN is the strongest external competitor, but the gap remains visible across the labelled sample-size range. The comparison supports the value of pattern-adaptive prediction rather than treating missing modalities only through fusion or reconstruction.

For the comparison, we use a residual-dependence setting with $q = 0.6$, where the shared representation remains independent but modality-specific representations become correlated through a common residual factor. We vary $\rho \in \{0, 0.3, 0.6\}$, which increases the calibration gap. The complete setting is provided in Supplementary Section E.2. We compare MOSAIC with Aligned-SPSM and four internal variants. By construction, Aligned-SPSM uses the same learned shared and modality-specific representations as MOSAIC, replacing the overlap-and-calibration head with a shared coefficient vector and sparse pattern-specific

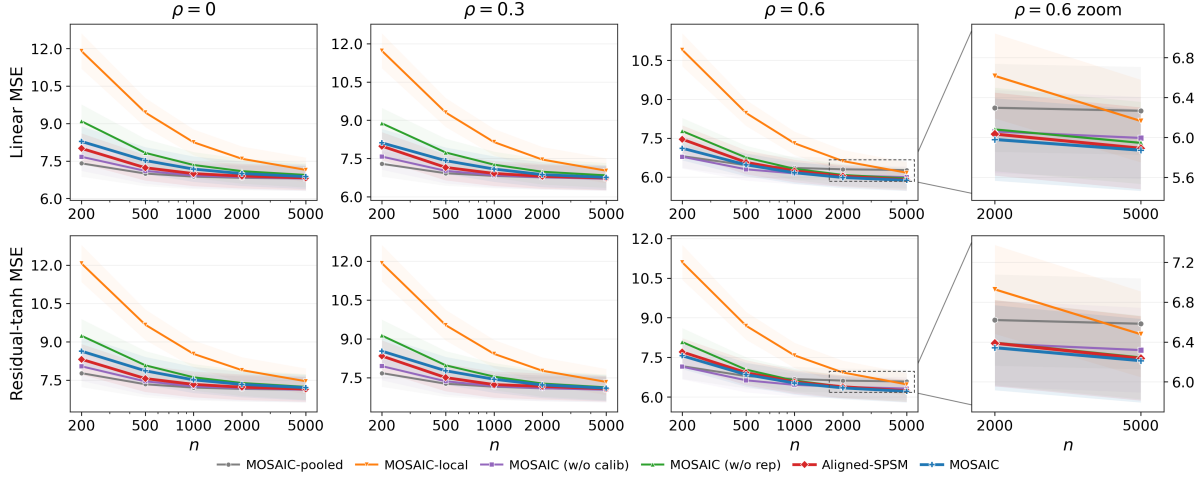


Figure 6: Downstream decomposition as residual dependence among modality-specific representations varies. Shaded bands are pointwise 95% confidence intervals for the mean over 100 replications. The right column enlarges the $\rho = 0.6$ results at $n \in \{2000, 5000\}$.

deviations. Its penalties are selected jointly by balanced-pattern validation MSE. MOSAIC-local uses only focal-pattern samples. MOSAIC-pooled ignores pattern structure by fitting one predictor on zero-filled representations from all patterns. MOSAIC (w/o calib) keeps the overlap-based preliminary predictor but removes the focal-pattern calibration step. MOSAIC (w/o rep) replaces the learned representations with modality-wise PCA features. The comparison with Aligned-SPSM shows a crossover rather than uniform dominance (Figure 6). Aligned-SPSM has lower mean MSE at every evaluated n when $\rho = 0$ and $\rho = 0.3$, although the difference becomes small at the largest n under $\rho = 0.3$. At $\rho = 0.6$, MOSAIC has lower mean MSE at every n in both observation regimes. This reversal is consistent with coefficient sharing being effective when pattern-specific discrepancies are small, whereas overlap-based prediction followed by focal-pattern calibration becomes advantageous as residual dependence and rule mismatch increase.

The internal variants provide complementary evidence. MOSAIC has lower mean MSE than MOSAIC-local and MOSAIC (w/o rep) in all evaluated settings. At $\rho = 0.6$, MOSAIC

overtakes MOSAIC-pooled from $n = 1000$ and MOSAIC (w/o calib) from $n = 2000$ in both observation regimes, while these simpler variants remain competitive at lower residual dependence or smaller sample sizes. Thus, the results support the value of overlap borrowing throughout, while the benefit of calibration emerges when both the rule mismatch and the focal-pattern sample size are sufficiently large. Additionally, we compare MOSAIC with the raw-imputation baseline and extend the analysis in Supplementary Section E.2.

5 Real Data Applications

We evaluate MOSAIC on next-visit intensive care unit (ICU) mortality prediction using the Medical Information Mart for Intensive Care (MIMIC) database ([Johnson et al. 2023](#)). MIMIC contains longitudinal structured codes, clinical notes, and medical images, but these modalities are not available for every patient. These naturally occurring modality-availability patterns arise from routine clinical workflows, providing a real-world blockwise missingness setting for evaluation. Additional details on cohort construction, modality-specific preprocessing, training protocol, dataset statistics, and empirical observed-modality-pattern distributions are provided in Supplementary Sections E.3 and E.5–E.8.

To examine performance under other tasks, we also evaluate MOSAIC under controlled modality masking on IEMOCAP emotion recognition with audio and text ([Busso et al. 2008](#)) and FairDomain glaucoma classification with two ophthalmic imaging modalities ([Tian et al. 2025](#)). Full results and analyses are reported in Supplementary Section E.9.

5.1 MIMIC Next-Visit ICU Mortality Prediction

We formulate the MIMIC application as a next-visit ICU mortality prediction task. To construct this task, we restrict the cohort to patients with at least two ICU visits, yielding 130,873 patients in total. We then split this cohort at the patient level into fixed training,

validation, and test sets in a 7 : 1 : 2 ratio. The validation set is used for model selection, and the test set is reserved for final evaluation. For patient i , let T_i denote the number of observed ICU visits; the covariate history consists of visits $1, \dots, T_i - 1$, and the outcome indicates whether mortality occurs at visit T_i . The final visit is used only to define the outcome and is excluded from the covariate history.

We construct covariates from three modalities. The structured-code covariates contain diagnosis, medication, diagnosis-related group (DRG), and laboratory event streams from historical visits. The image covariates contain a chest X-ray when available ([Johnson et al. 2024](#)). The note covariates contain discharge notes and radiology notes from historical visits. A modality is coded as missing if no historical information from that modality is available.

In addition to overall test performance, we report performance within observed-modality patterns to assess whether the observed gains persist across the prediction tasks induced by different modality-availability patterns. To avoid noisy pattern-level estimates, we display only patterns with at least 20 test samples and at least one positive test case. The retained MIMIC patterns include code-only, code+note, and code+note+image configurations; detailed sample sizes for each observed-modality pattern are reported in Supplementary Section E.8.

We compare MOSAIC with the comparison methods introduced in Section 4.2, together with the pattern-submodel baseline PatternSub ([Fletcher Mercaldo & Blume 2020](#)). Figure 7 reports overall and pattern-specific test performance over five random seeds.

MOSAIC achieves the highest overall mean area under the receiver operating characteristic curve (AUROC), area under the precision–recall curve (AUPRC), and F1 score among the evaluated methods, with values of 0.861, 0.238, and 0.326, respectively. In the code-only pattern, the separation between MOSAIC and the comparison methods is generally larger than in the other modality-availability patterns. This behaviour is consistent with

cross-pattern borrowing followed by focal-pattern correction: observations from richer patterns can contribute to the code-only predictor through the shared representation and the code-specific representation, both of which are available from code histories. In the code+note+image pattern, MOSAIC again has the highest mean AUROC, AUPRC, and F1, showing that its advantage is retained when all three modalities are observed. By contrast, the leading methods are more closely grouped in the code+note pattern, where the advantage of MOSAIC is less pronounced than in the other two patterns. Overall, MOSAIC performs favourably across heterogeneous modality-availability patterns, although the magnitude of its advantage varies with the observed modalities.

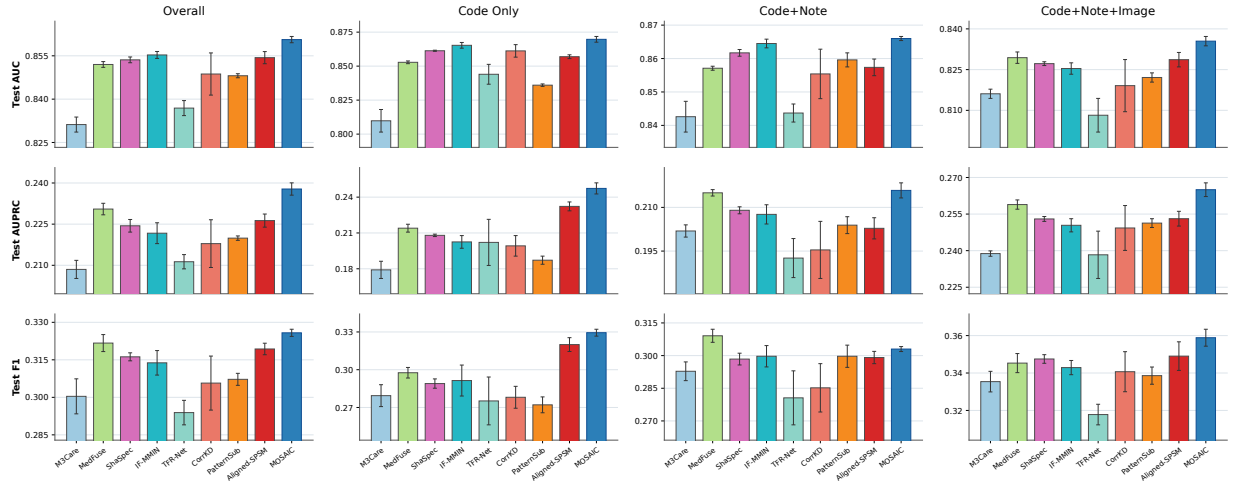


Figure 7: Performance on MIMIC next-visit ICU mortality prediction. Bars show mean performance over five seeds, with error bars indicating standard deviations.

5.2 Component Analysis on the MIMIC Application

We further compare MOSAIC with the four internal variants from Section 4.2 to isolate the contributions of cross-pattern borrowing, representation learning, calibration, and pattern-specific final prediction.

Figure 8 reports the component analysis. The full model achieves the best performance

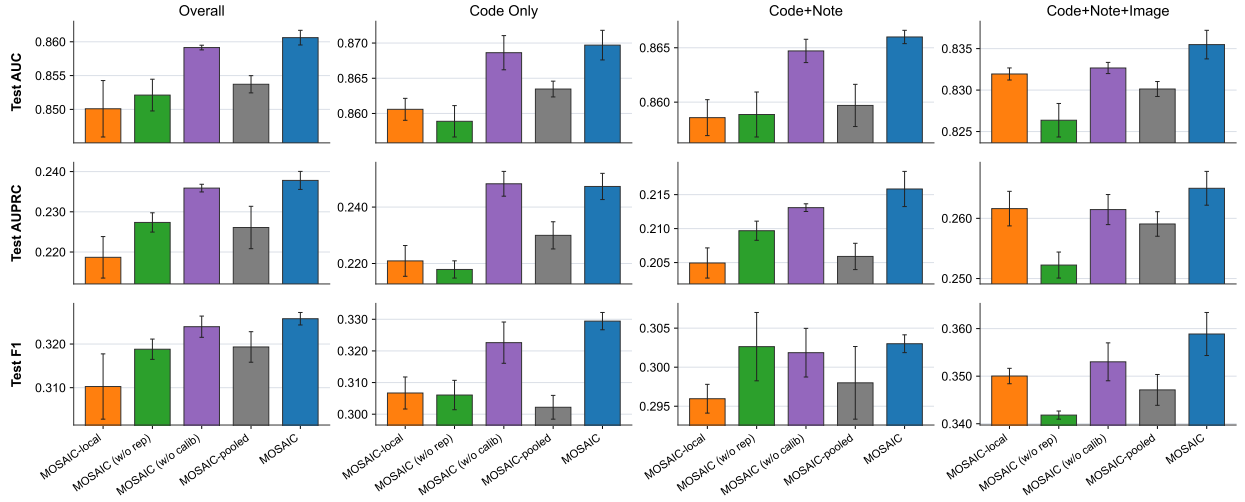


Figure 8: Component analysis of MOSAIC on MIMIC next-visit ICU mortality prediction. MOSAIC-local removes cross-pattern borrowing, MOSAIC (w/o rep) removes the representation-learning component, MOSAIC (w/o calib) removes focal-pattern calibration, and MOSAIC-pooled removes pattern-specific final prediction. Bars show mean performance over five random seeds, with error bars indicating standard deviations.

and also leads in most pattern-specific comparisons, although the size of the differences varies across availability patterns. Its gains over MOSAIC (w/o rep) highlight the value of the learned shared and modality-specific representations, while its gains over MOSAIC (w/o calib) indicate that focal-pattern correction provides an additional refinement. The largest differences between the full model and MOSAIC-local or MOSAIC-pooled occur in the code-only group. With neither notes nor images available, a local predictor cannot use observations from patterns with additional modalities. The comparison with MOSAIC-local reflects the value of cross-pattern borrowing, while the comparison with MOSAIC-pooled shows the benefit of pattern-specific final prediction.

The improvement of the full model over MOSAIC-local is smaller in the code+note+image group. This is expected, as this group already contains the richest observed information, leaving less scope for borrowing from other patterns to improve prediction. In the code+note

pattern, the differences between MOSAIC and its components are more modest. This is consistent with the pattern-specific results in Figure 7, where several strong methods have similar performance in this observed-modality configuration.

6 Conclusion

MOSAIC borrows information across multimodal missingness patterns while retaining a separate prediction rule for each pattern. Once modalities are aligned using co-observed data, observations from patterns with no raw modality in common can still contribute through the shared representation, while focal-pattern calibration accounts for the remaining rule discrepancy. The theory and experiments show that the gain over local fitting depends on representation coverage, the pattern-specific sample size, and the size of the required correction. The current guarantees rely on a linear latent-representation model and MCAR; extending them to nonlinear representations and more general missingness mechanisms remains open.

References

- Argelaguet, R., Velten, B., Arnol, D., Dietrich, S., Zenz, T., Marioni, J. C., Buettner, F., Huber, W. & Stegle, O. (2018), ‘Multi-omics factor analysis—a framework for unsupervised integration of multi-omics data sets’, *Molecular Systems Biology* **14**(6), e8124.
- Baltrušaitis, T., Ahuja, C. & Morency, L.-P. (2019), ‘Multimodal machine learning: A survey and taxonomy’, *IEEE Transactions on Pattern Analysis and Machine Intelligence* **41**(2), 423–443.
- Bertsimas, D., Delarue, A. & Pauphilet, J. (2025), ‘Adaptive optimization for prediction with missing data’, *Machine Learning* **114**, 124.
- Busso, C., Bulut, M., Lee, C.-C., Kazemzadeh, A., Mower, E., Kim, S., Chang, J. N., Lee, S. & Narayanan, S. S. (2008), ‘Iemocap: interactive emotional dyadic motion capture database’, *Language Resources and Evaluation* **42**(4), 335–359.

- Chartsias, A., Joyce, T., Giuffrida, M. V. & Tsafaris, S. A. (2018), ‘Multimodal mr synthesis via modality-invariant latent representation’, *IEEE Transactions on Medical Imaging* **37**(3), 803–814.
- Fletcher Mercaldo, S. & Blume, J. D. (2020), ‘Missing data and prediction: the pattern submodel’, *Biostatistics* **21**(2), 236–252.
- Hayat, N., Geras, K. J. & Shamout, F. E. (2022), Medfuse: Multi-modal fusion with clinical time-series data and chest x-ray images, *in* ‘Machine Learning for Healthcare Conference’, PMLR, pp. 479–503.
- Hazarika, D., Zimmermann, R. & Poria, S. (2020), Misa: Modality-invariant and -specific representations for multimodal sentiment analysis, *in* ‘Proceedings of the 28th ACM International Conference on Multimedia’, pp. 1122–1131.
- Jin, Y. & Rothenhäusler, D. (2023), ‘Modular regression: Improving linear models by incorporating auxiliary data’, *Journal of Machine Learning Research* **24**(351), 1–52.
- Johnson, A. E., Bulgarelli, L., Shen, L., Gayles, A., Shammout, A., Horng, S., Pollard, T. J., Hao, S., Moody, B., Gow, B. et al. (2023), ‘MIMIC-IV, a freely accessible electronic health record dataset’, *Scientific Data* **10**(1), 1.
- Johnson, A., Lungren, M., Peng, Y., Lu, Z., Mark, R., Berkowitz, S. & Horng, S. (2024), ‘MIMIC-CXR-JPG - chest radiographs with structured labels’, *PhysioNet*. Version 2.1.0.
- Kline, A., Wang, H., Li, Y., Dennis, S., Hutch, M., Xu, Z., Wang, F., Cheng, F. & Luo, Y. (2022), ‘Multimodal machine learning in precision health: A scoping review’, *npj Digital Medicine* **5**(1), 171.
- Krones, F., Marikkar, U., Parsons, G., Szmul, A. & Mahdi, A. (2025), ‘Review of multimodal machine learning approaches in healthcare’, *Information Fusion* **114**, 102690.
- Le Morvan, M., Josse, J., Moreau, T., Scornet, E. & Varoquaux, G. (2020), Neumiss networks: Differentiable programming for supervised learning with missing values, *in* ‘Advances in Neural Information Processing Systems’, Vol. 33.
- Le Morvan, M., Prost, N., Josse, J., Scornet, E. & Varoquaux, G. (2020), Linear predictor on linearly-generated data with missing values: Non consistency and solutions, *in* ‘Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics’, Vol. 108 of *Proceedings of Machine Learning Research*, PMLR, pp. 3165–3174.

- Lee, M. & Pavlovic, V. (2021), Private-shared disentangled multimodal vae for learning of latent representations, *in* ‘Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)’, pp. 1692–1700.
- Li, M., Yang, D., Zhao, X., Wang, S., Wang, Y., Yang, K., Sun, M., Kou, D., Qian, Z. & Zhang, L. (2024), Correlation-decoupled knowledge distillation for multimodal sentiment analysis with incomplete modalities, *in* ‘Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition’, pp. 10353–10363.
- Ma, T., Wang, T. & Samworth, R. J. (2025), ‘Deep learning with missing data’, *arXiv preprint arXiv:2504.15388*.
- Sell, T., Berrett, T. B. & Cannings, T. I. (2024), ‘Nonparametric classification with missing data’, *The Annals of Statistics* **52**(3), 1178–1200.
- Shi, Y., Siddharth, N., Paige, B. & Torr, P. H. S. (2019), Variational mixture-of-experts autoencoders for multi-modal deep generative models, *in* ‘Advances in Neural Information Processing Systems’, Vol. 32.
- Stempfle, L., Panahi, A. & Johansson, F. D. (2023), Sharing pattern submodels for prediction with missing values, *in* ‘Proceedings of the AAAI Conference on Artificial Intelligence’, Vol. 37, pp. 9882–9890.
- Tian, Y., Wen, C., Shi, M., Afzal, M. M., Huang, H., Khan, M. O., Luo, Y., Fang, Y. & Wang, M. (2025), Fairdomain: Achieving fairness in cross-domain medical image segmentation and classification, *in* A. Leonardis, E. Ricci, S. Roth, O. Russakovsky, T. Sattler & G. Varol, eds, ‘Computer Vision – ECCV 2024’, Springer Nature Switzerland, Cham, pp. 251–271.
- Tran, L., Liu, X., Zhou, J. & Jin, R. (2017), Missing modalities imputation via cascaded residual autoencoder, *in* ‘Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)’, pp. 1405–1414.
- Wang, H., Chen, Y., Ma, C., Avery, J., Hull, L. & Carneiro, G. (2023), Multi-modal learning with missing modality via shared-specific feature modelling, *in* ‘Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)’, pp. 15878–15887.
- Wu, M. & Goodman, N. (2018), Multimodal generative models for scalable weakly-supervised learning, *in* ‘Advances in Neural Information Processing Systems’, Vol. 31.

- Wu, R., Wang, H., Chen, H.-T. & Carneiro, G. (2026), ‘Deep multimodal learning with missing modality: A survey’, *Transactions on Machine Learning Research* . Survey Certification.
- Yu, G., Li, Q., Shen, D. & Liu, Y. (2020), ‘Optimal sparse linear prediction for block-missing multi-modality data without imputation’, *Journal of the American Statistical Association* **115**(531), 1406–1419.
- Yuan, Z., Li, W., Xu, H. & Yu, W. (2021), Transformer-based feature reconstruction network for robust multimodal sentiment analysis, *in* ‘Proceedings of the 29th ACM International Conference on Multimedia’, pp. 4400–4407.
- Zhang, C., Chu, X., Ma, L., Zhu, Y., Wang, Y., Wang, J. & Zhao, J. (2022), M3care: Learning with missing modalities in multimodal healthcare data, *in* ‘Proceedings of the 28th ACM SIGKDD conference on knowledge discovery and data mining’, pp. 2418–2428.
- Zhao, J., Li, R. & Jin, Q. (2021), Missing modality imagination network for emotion recognition with uncertain missing modalities, *in* ‘Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)’, Association for Computational Linguistics, Online, pp. 2608–2618.
- Zuo, H., Liu, R., Zhao, J., Gao, G. & Li, H. (2023), Exploiting modality-invariant feature for robust multimodal emotion recognition with missing modalities, *in* ‘ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)’, pp. 1–5.